



Optimising performances of LoRa based IoT enabled wireless sensor network for smart agriculture

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ABSTRACT

The significance of maintaining food security is particularly emphasized in regions such as Malaysia, where there is a substantial reliance on imports of fruits and vegetables. The local farming industry requires advancements in technology to address the complexities of food production. This motivates the exploration of implementing Internet of Things (IoT) solutions to reduce reliance on manpower and enhance farming efficiency. However, the current integrated IoT solutions for smart farms can be expensive and complicated, rendering them redundant for small-area farms that do not require advanced technologies or machine learning. This study aims to optimize IoT LoRa parameter settings, focusing on providing affordable solutions tailored for fruit and vegetable farms. The main goal is to address the common issue faced by farmers—investing in expensive IoT-enabled wireless sensors without realizing a return on investment. To overcome this challenge, a simple and cost-effective IoT solution is proposed, utilising sensors and readily available applications to achieve efficient farming practises. The main objective is to develop an integrated IoT solution tailored for fruit and vegetable farms, facilitating improved productivity and growth. The solution incorporates sensors to monitor and control various aspects of crop cultivation, collecting essential data. The research development here also aims to inspire low-cost IoT solutions suitable for small-scale food plantation farms by using the long-range physical communication technique (LoRa) protocol. Leveraging IoT technologies in a simplified manner allows local farming industries to boost productivity, lower operational costs, and effectively address food security concerns. In addition to the IoT technology development for smart farms, the research here also investigates the various factors affecting the performance of LoRa-enabled IoT solutions for smart farms, including LoRa frequency, distance, environment, and the performance of the IoT system under different weather conditions, showed a significant effect on system performance ($p \leq 0.05$). This comprehensive approach aims to contribute valuable insights for the development and implementation of cost-effective IoT solutions tailored for small-scale agricultural practices. In conclusion, this project demonstrates the efficiency of LoRa-based IoT in agriculture and highlights areas for improvement. Future directions involve improving scalability, streamlining the user interface, and adding sophisticated features to support the development of productive, networked, and environmentally friendly farming methods.

1. Introduction

COVID-19 and the Russia-Ukraine war have led to food insecurity in import-dependent countries, highlighting the critical importance of global food security [1]. The agriculture industry faces significant challenges, including population growth, limited cultivable land, and the impacts of global warming, which have resulted in a decline in global food production [2]. The projected global population of 8.5 billion by 2030 and 9.7 billion by 2050 necessitates urgent development of sustainable food production solutions and measures to mitigate the

effects of climate change [2]. Smart agriculture, driven by cutting-edge technologies such as the Internet of Things (IoT) and real-time monitoring through sensors, offers promising opportunities to enhance agricultural productivity and address these challenges. Integrating IoT into farming practices enables farmers to closely monitor plantation areas, access statistical data through various devices, and implement farm security solutions for crop protection. This enables them to promptly respond to emerging issues and mitigate risks, ensuring sustainable food production for the future.

The advent of Industry 4.0 has revolutionised industries by

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incorporating advanced technologies such as IoT, Artificial Intelligence (AI), Augmented Reality (AR), real-time communication, big data, remote sensing, and monitoring and control [3,4]. In the agricultural sector, smart farming has emerged as a transformative approach by integrating Industry 4.0 technologies [5]. As the world's population grows at an accelerated rate, it is now imperative to ensure sustainable food production. Numerous solutions that have been developed using the Internet of Things are intended to mitigate the difficulties that farmers encounter, including problems associated with the production and storage of food.

Adopting the sustainable development goals commits the global community to eradicating poverty, ending hunger, and taking immediate action to address climate change and its effects by 2030 [6]. Besides that, the decrease in the profitability of farms has led young people to abandon agriculture, with the disappearance of small and medium-sized farms, negative effects and environmental impacts, and the disappearance of rural communities [7].

The adoption of IoT has become crucial for implementing smart farming practises, particularly in remote areas. It facilitates seamless data flow between backhaul and edge computing systems, connecting sensors, actuators, and other on-the-ground devices [8]. Communication technology choices are pivotal in ensuring effective data transmission and communication in smart farming. Extensive scientific research has been conducted to explore communication techniques and protocols for smart farming, focusing on networking protocols, energy efficiency, robustness, scalability, and more [9]. Commonly used IoT technologies include low-power Wi-Fi, Bluetooth Low Energy (BLE), DASH7 Alliance Protocol (D7A), Long Range (LoRa), and Long-Range Wide Area Network (LoRaWAN) [9]. For long-distance wireless connectivity, technologies such as cellular networks (2G, 3G, 4G, and 5G) and Low-power Wide Area Networks (LPWANs) are widely employed, with LPWANs being specifically designed for long-range communication with low power requirements. IoT modules based on the LoRaWAN protocol are expected to meet specific requirements such as low cost, low power consumption, a small form factor, rapid prototyping, and ease of programming (compatibility with the Arduino environment) [7].

This paper introduces an integrated IoT solution customised for fruit and vegetable farms with the goal of enhancing productivity and addressing challenges faced by the local farming industry. The solution incorporates sensor-based monitoring and control mechanisms to effectively manage various aspects of crop cultivation, enabling seamless data collection (see Table 1 and 2). Additionally, the research investigated the effectiveness of the deployment of LoRa technologies for sensor data transmission through IoT gateways. Here, it was specifically examined how elements like LoRa frequency, distance, environment, and weather conditions affect the performance of LoRa-enabled IoT solutions for smart farms. The agriculture sector can achieve increased productivity, decreased operational costs, and efficient methods to maintain food security by embracing simplified IoT technologies made possible by LoRa.

Therefore, this study aims to make the goals clear. The main goal is to optimize IoT LoRa parameter settings to provide low cost IoT solutions and get better outcomes. Purchasing pricey wireless sensors with Internet of Things capabilities and not seeing a return on investment is a problem that many farmers frequently encounter. In response this paper offers a solution by providing smaller-scale farmers with a cost-effective idea. The calibration of a less expensive IoT system customized to meet these farmers' demands is part of the study.

2. Materials

2.1. Circuit hardware & software components

2.1.1. Sensors

Table 1 and 2 highlight the main functions and specifications for the sensor systems adopted in this study.

Table 1

Main function for all sensor systems.

No	Sensor	Function
1	DHT22 temperature and humidity sensor	To measure and monitor environmental conditions [10].
2	FC-28 Soil Moisture Sensor	To measure the moisture content of soil and similar materials [11].
3	Raindrop Sensor	To detect the presence of rain in the agriculture sector [12].
4	I2C Liquid Crystal Display (LCD)	A 16×2 I2C LCD Display to present information such as rainfall, soil moisture level, temperature, and humidity [13].
5	DS3231 real-time clock (RTC) module	To provide accurate timekeeping and timestamping functionality [14]. In this project, control the time for the transmitter node to send data simultaneously.

Table 2

Specifications for all sensor systems.

No	Sensor	Specifications
1	DHT22 temperature and humidity sensor	Measurement Range: $-40\text{ }^{\circ}\text{C}$ – $80\text{ }^{\circ}\text{C}$ for temperature, 0%–100% for humidity Accuracy: $\pm 0.5\text{ }^{\circ}\text{C}$ for temperature, $\pm 2\%$ for humidity
2	FC-28 Soil Moisture Sensor	Operating Voltage: 3.3 V–5 V Detection Range: Adjustable based on soil moisture levels
3	Raindrop Sensor	Operating Voltage: 3.3 V–5 V Detects raindrops and provides analog output
4	I2C Liquid Crystal Display (LCD)	Communication: I2C Display: 16×2 character LCD
5	DS3231 real-time clock (RTC) module	Accuracy: ± 2 ppm from $0\text{ }^{\circ}\text{C}$ to $40\text{ }^{\circ}\text{C}$ Integrated Temperature-Compensated Crystal Oscillator (TCXO)

2.1.2. LORA gateway

The LoRa gateway plays a crucial role in facilitating data transmission from the transmitter node to the receiver node. Due to the farm's size and lack of internet access, LoRa has been deployed as a necessary part for dependable data transfer from the system unit to the reception unit. The data is subsequently transmitted from the receiver unit to an IoT cloud platform, Thingspeak, for safe data management and storage. The issues faced by the farm's extensive covering are overcome using LoRa technology, which permits long-range communication using radio frequency technology.

LoRa technology has a number of benefits that make it ideal for wireless data transfer in Internet of Things applications. Data can be transmitted over great distances thanks to its long-range capabilities, which makes it perfect for smart farms with extensive coverage. Additionally, because LoRa uses so little power, energy is used efficiently, extending the useable life of battery-powered devices. Additionally secure, LoRa data transmission ensures the secrecy and integrity of the transmitted data. To efficiently encode and transmit data, this technique uses multi-symbol data formats and chirp spread spectrum (CSS) modulation. A smooth integration into the cloud environment is made possible using LoRa, which enables connections and communication between sensors, gateways, electronics, and machines. According on the region, different frequency bands may be used by LoRa, such those between 433 and 435 MHz in Malaysia and 919 and 923 MHz in Asia [15].

2.1.3. IoT systems

Table 3 highlights the specifications and functions for all IoT systems adopted in this study.

2.2. NodeMCU ESP32

The NodeMCU ESP32 exhibits remarkable capabilities for IoT

Table 3

Specifications and functions for all IoT systems.

No	IoT	Specifications and Functions
1	NodeMCU ESP32	Microcontroller: ESP32 CPU: Dual-core Tensilica LX6 Connectivity: Wi-Fi, Bluetooth Digital I/O Pins: 36 Analog Input Pins: 18 Function: IoT device development, wireless connectivity, supports Arduino IDE.
2	Arduino Uno R3	Microcontroller: ATmega32P CPU: 8-bit AVR Digital I/O Pins: 14 Analog Pins: 6 Function: General-purpose microcontroller for various electronics projects, programming with Arduino IDE.
3	ThingSpeak IoT Platform	Cloud-based IoT platform Features: Data collection, analysis, visualisation Functions: Allow IoT device data to be uploaded, stored, analyzed, and visualized in real-time through web-based interfaces.
4	Blynk Mobile Application	Mobile app for IoT control and monitoring Features: Drag-and-drop UI, able to customization Functions: Allows users to create interfaces for controlling and monitoring IoT devices through a smart-phone app.

applications and was chosen as the receiver microcontroller in the smart farm system. The system's dual-core Xtensa LX6 processor, which has a maximum clock speed of 240 MHz, has a lot of processing power and can handle time-sensitive data-intensive operations [16]. The ESP32 enables smooth connectivity to wireless networks and compatible devices with support for 2.4 GHz Wi-Fi and Bluetooth 4.2/BLE, facilitating effective data transfer and communication. For programme execution and data storage, it has vast memory resources, including up to 520 KB of SRAM and 4 MB of Flash memory. The Arduino IDE, powered by a growing open-source community, makes programming easier while the GPIO pins provide flexibility in incorporating external devices and sensors. Overall, the NodeMCU ESP32 proves to be a flexible and effective microcontroller, ideal for implementing the receiver functionality in the smart farm system.

2.3. Arduino UNO R3

The Atmega328P microcontroller in the Arduino Uno, a well-liked open-source development board, was selected as the microcontroller for the transmitter node in the smart farm system [17]. It offers digital and analog input and output pins for sensor integration and data processing. With its compact size, low power consumption, and versatility, the Arduino Uno plays a crucial role in collecting, processing, and transmitting data for effective farm monitoring and crop cultivation optimisation.

2.4. Thingspeak IoT platform

The ThingSpeak IoT platform, renowned for its real-time data visualisation and analysis capabilities, was integrated into the smart farm system. Using an IoT device with LoRa technology, a crop field monitoring system was developed, with data collected and transmitted from the transmitter node to the receiver node. The receiver node, powered by the ESP32, facilitated the connection between the smart farm system and the ThingSpeak platform. As a pivotal component, ThingSpeak securely stored the collected sensor data in the cloud, serving as a dependable backup for further analysis and monitoring.

2.5. Blynk mobile application

Using IOS or Android smartphones, users can manage Arduino, Raspberry Pi, and NodeMCU gadgets using Blynk. By giving the right

addresses, it provides a graphical user interface (GUI) or human-machine interface (HMI) that can be configured. Blynk, which was created especially for the Internet of Things, offers several capabilities including remote hardware control, real-time sensor data presentation, data storage and visualisation, and others. The Blynk mobile application will be used in the context of this project to display sensor data such as motion, rain presence, temperature, humidity, and soil moisture. The Blynk mobile application will give users access to real-time sensor data from the smart farm, allowing them to keep tabs on the farm's circumstances. The mobile application will also provide users the opportunity to turn on and off the IoT device and receive warnings when motion is detected.

3. Methods

3.1. Proposed system

This journal paper introduces a real-time monitoring system for smart farms that is both effective and affordable. The proposed system is based on the wireless sensor network protocol for Internet of Things (IoT) applications and uses long-range radio technology. The technology specifically makes use of LoRa, a wireless platform created by Semtech that allows farmers to efficiently manage their irrigation systems on big fields without the need for an internet connection. The unique quality of LoRa is its broad coverage area, which enables data transfer over long distances and qualifies it for wide-area communication in rural areas. The significance of LoRaWAN, an up-and-coming communication technology that establishes the communication protocol and architecture for LoRa-based Internet of Things networks, is also highlighted in the article. To increase the overall dependability and efficiency of the proposed system, LoRaWAN ensures secure communication, scalability, and interoperability across devices from multiple vendors.

The transmitter node with sensors and an Arduino Uno, LoRa communication, and the receiver node with the ESP32 are shown as the system's building elements in Fig. 1. For data storage and user-friendly data visualisation, the proposed system makes use of the Thingspeak cloud storage platform and the Blynk mobile application.

Fig. 2 presents the flowchart of a plant monitoring system for smart farms, consisting of the sensing module, IoT platform, and mobile applications. The sensing module comprises the transmitter node and receiver node, each equipped with a microcontroller for measuring farm conditions. The IoT platform serves as a gateway, collecting sensor measurements from the transmitter node, grouping them into data packets, and transmitting them wirelessly to the receiver node via LoRa communication. The receiver node reads the data and pushes it to the cloud platform for further processing. Utilising cloud technology and mobile applications, such as Blynk, enables remote monitoring and control of crop field conditions, facilitating efficient data exchange and the resolution of plantation issues through IoT technology.

Fig. 3 shows the proposed system architecture for the project. The proposed IoT-based smart farm project includes a lot of main components, including a system unit, LoRa, an IoT Cloud Platform, and a Blynk Application. These main components are connected and interact with each other by radio frequency, wireless, and the internet.

The layout design for placing the system on a fruit farm consists of four key components: the transmitter node, receiver node, solar power system, and wire connections. The transmitter node is strategically positioned among the fruit trees to effectively monitor environmental conditions and ensure accurate data collection. To sustain the transmitter node's power, a solar power system is installed in the centre of the farm, utilising solar panels strategically placed for optimal sunlight exposure and providing a renewable source of electricity. The receiver node is located in the office or control centre of the farm, enabling easy access to the received data for analysis, storage, and further processing. Wire connections are established to facilitate seamless communication between the components, considering factors like cable length,

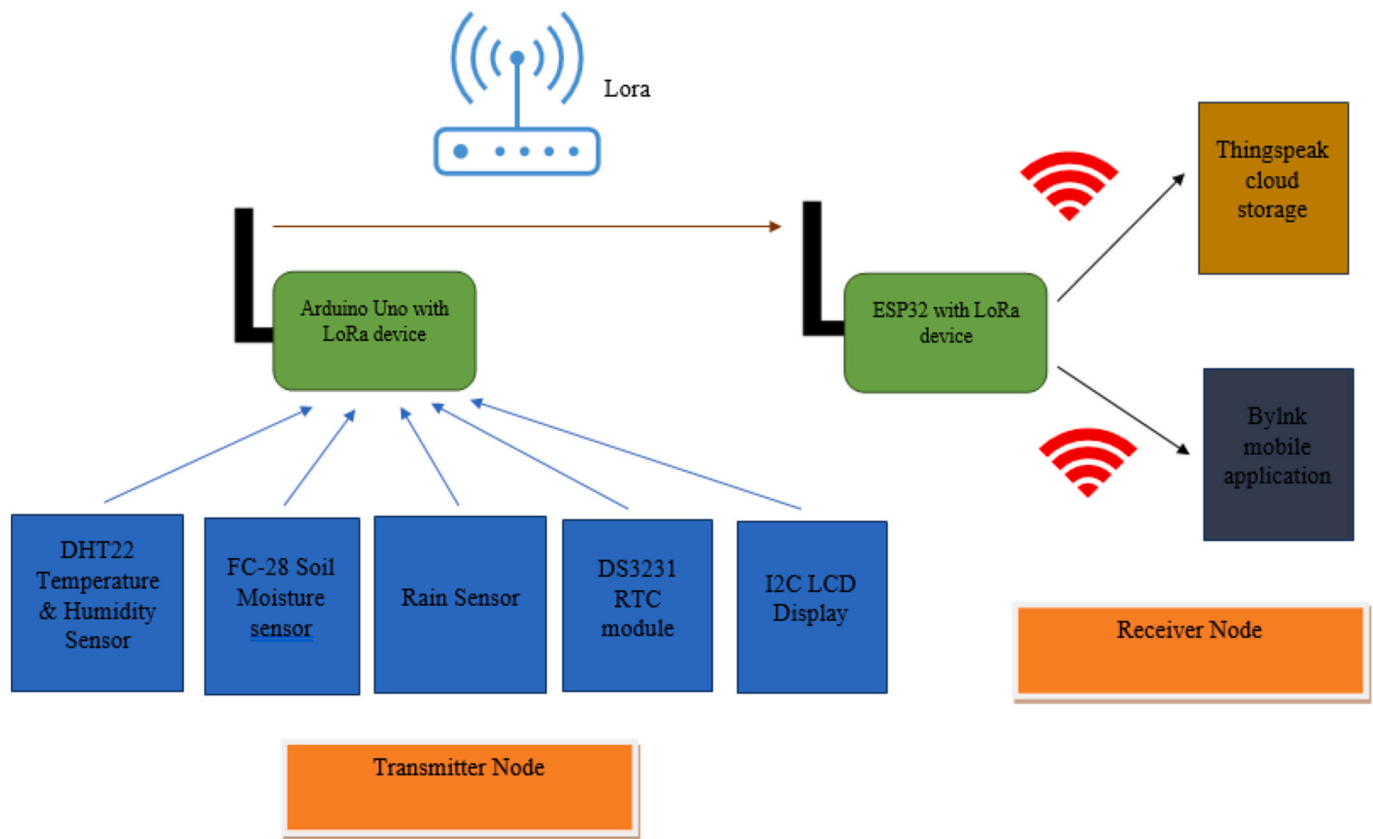


Fig. 1. The block diagram of the system.

durability, and environmental protection. Fig. 4 visually illustrates the layout design, offering a clear depiction of the component arrangements within the fruit farm system.

The transmitter node sensing system architecture was designed to efficiently collect, display, and transmit crucial data from multiple sensors in the smart farm. The system utilised an Arduino Uno microcontroller as the central control unit, enabling seamless integration and processing of data from various sensors. Connected directly to the microcontroller, the DHT22 sensor measured temperature and humidity, the raindrop sensor detected rainfall patterns, the FC-28 soil moisture sensor measured soil moisture content, and the DS3231 real-time clock module ensured precise timekeeping. An I2C LCD display provided a user-friendly interface for real-time monitoring of environmental conditions. To enable wireless data transmission, the Arduino Uno was connected to a LoRa module, facilitating secure and reliable communication with the receiver node or central server. Fig. 5 visually represents the block diagram of the transmitter node sensing system, illustrating the interconnections between components for efficient data collection and transmission in the smart farm.

The integrated IoT solution for smart farms relies heavily on the receiver node system to efficiently gather, process, and access the data broadcast from the transmitter node system. The robust ESP32 microcontroller, which is linked to a LoRa module for long-distance wireless communication, lies at the heart of the system architecture. The ESP32 analyses the sensor values before safely uploading the data to the ThingSpeak cloud platform for archival, visualisation, and analysis. The ESP32 is also linked to the Blynk mobile application, which offers a user-friendly interface for remote access, control, and real-time monitoring of the smart farm system. With the help of this complete system design, farmers and operators can manage their farms effectively and make data-driven decisions. The connected parts that are in charge of data reception and processing in the smart farm are shown in Fig. 6 block diagram of the receiver system.

3.2. Circuit diagram and schematic

Fig. 7 shows the circuit connection for the transmitter node. A power bank source that is attached to the USB port of the Arduino Uno microcontroller powers the entire transmitter node system. A number of sensors and modules are linked to the Arduino Uno, which acts as the system's central hub. These include the DHT22 sensor for monitoring temperature and humidity, the FC-28 soil moisture sensor, the DS3231 RTC module for precise timekeeping, the I2C LCD 16 × 12 display, and the raindrop sensor. With this setup, the Arduino Uno can gather data from these sensors and make it easier for it to be transmitted to the receiver node in the smart farm system.

Fig. 8 depicts the circuit connection for the receiver node. The LoRa-02 module and the ESP32 microcontroller are both powered by a 5V1A power source. The ESP32 and LoRa-02 module are coupled, which makes it easier to receive data from the transmitter node. This configuration ensures the proper functioning and data transfer capabilities of the receiver node in the smart farm system.

3.3. Experimental flow chart design

In Fig. 9, the experimental flow chart in this study comprises five sets of experiments aimed at evaluating different aspects of the integrated IoT solution for smart agriculture.

The first set focuses on determining the optimal frequency signal for LoRa communication between the transmitter node and receiver node. Outdoor tests are conducted at three different frequencies: 433 Hz, 866 Hz, and 915 Hz. The objective is to assess the performance of LoRa communication at these frequencies and identify the most effective one. During this set of experiments, signal-to-noise ratio (SNR) and received signal strength indicator (RSSI) are measured as performance metrics.

The second set of experiments aims to determine the maximum distance that LoRa signals can cover from the transmitter node to the

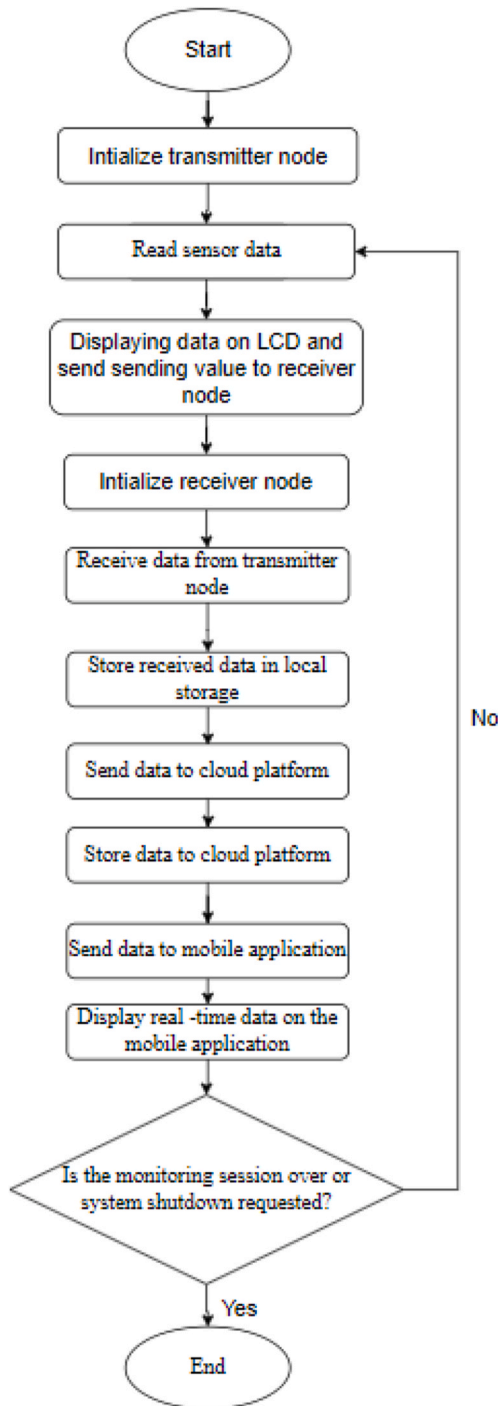


Fig. 2. Project methodology flowchart.

receiver node. Different distances are tested to evaluate the signal strength and coverage range. This information is crucial to understanding the communication capabilities and limitations of the IoT solution.

In the third set of experiments, known as the LoRa Environment series, the performance of LoRa communication is tested in various locations, including single-story housing, double-story housing, high-rise buildings, and farms. This assessment helps evaluate the impact of different environments on signal quality and reliability. SNR and RSSI measurements are again utilised to assess the performance under different conditions.

The fourth set of experiments focuses on analysing the IoT solution's

performance under different weather conditions. Parameters such as soil moisture, humidity, temperature, and rainfall are measured to understand the system's resilience in adverse weather. By examining how weather affects data transmission and sensor readings, the IoT solution's suitability for real-world farm environments can be determined.

The fifth and final set of experiments involves testing the performance of multiple transmitter nodes with a single receiver node. Sequential data transmission over a 1-h period is evaluated to assess the system's reliability and consistency. This test provides insights into the IoT solution's ability to handle simultaneous data transmission from multiple sources.

Throughout these experiments, data collection and analysis will be conducted to measure and evaluate the performance of the integrated IoT solution for smart agriculture. The results obtained from each experiment set will offer valuable insights into the optimal frequency, distance coverage, environmental factors, weather resilience, and multi-node performance. These findings will contribute to the development of a robust and efficient IoT solution specifically tailored for smart farms.

4. Result and discussion

4.1. LoRa frequency tests

The experimental setup for the LoRa frequency test involved conducting tests in an empty field located in Kidurong, Bintulu, Sarawak. The transmitter node was placed at distances of 50 and 100 m, and the receiver node (gateway module) was positioned at a fixed location in the same field. The latitude and longitude of the receiver node were recorded as 3.23148 and 113.07527, respectively. RSSI and SNR values were measured for different frequencies, with three interval transmissions recorded for each frequency. The LoRa communication range between the transmitter node and receiver devices spanned from 50 m to 100 m. Both the transmitter node and receiver node LoRa modules were configured with identical parameters to ensure consistent outdoor coverage. For the most efficient and low-energy usage of LoRa device setup parameters, refer to detailed information available in Table 4 [18].

The experimental setup involved monitoring RSSI and SNR values using a laptop's serial monitor and the Blynk app on a smartphone, as shown in Fig. 10. The recorded data provided insights into signal transmission quality at different frequencies (433 Hz, 866 Hz, and 915 Hz) and distances (100 m and 50 m).

Based on the results presented in Table 5 the optimal frequency for LoRa communication between the receiver node and transmitter node was found to be 433 Hz. At 50 m, the average Packet Signal Strength Indicator (PSSI) at 433 Hz was significantly higher (−92 dBm) compared to both 866 Hz (−121 dB m, $p \leq 0.01$) and 915 Hz (−121 dB m, $p \leq 0.01$). This superiority in PSSI values at 433 Hz persisted even at 100 m, with an average PSSI of −112 dB m, demonstrating a statistically significant improvement over 866 Hz (−122 dB m, $p \leq 0.01$) and 915 Hz (−121 dB m, $p \leq 0.01$).

Moreover, the Signal-to-Noise Ratio (SNR) at 433 Hz exhibited notable advantages. At 50 m, the average SNR was 6.50 dB, presenting a statistically significant enhancement over 866 Hz (−21.58 dB, $p \leq 0.01$) and 915 Hz (−21.58 dB, $p \leq 0.01$). At 100 m, the average SNR at 433 Hz remained higher (−7.17 dB) compared to 866 Hz (−22.67 dB, $p \leq 0.01$) and 915 Hz (−24 dB, $p \leq 0.01$). Additionally, 433 Hz exhibited higher Signal-to-Noise Ratio (SNR) values, indicating a lower probability of errors in received packets and better communication performance. Moreover, using a LoRa transmission frequency of 433 MHz offers the advantage of low energy consumption, making it suitable for power-efficient operations in smart farming applications [19].

4.2. LoRa distance tests

The LoRa Distance Test was conducted along Jalan Pangeran Matusin, Kidurong, Bintulu, Sarawak, to determine the maximum

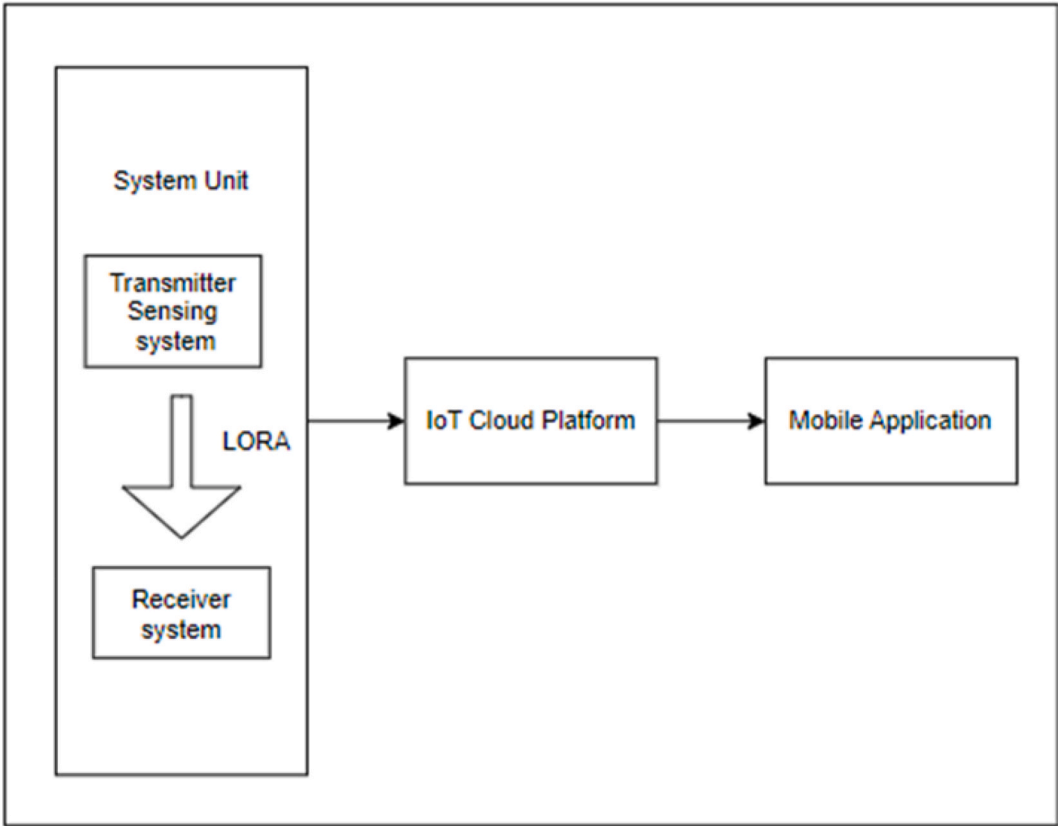


Fig. 3. System architecture.

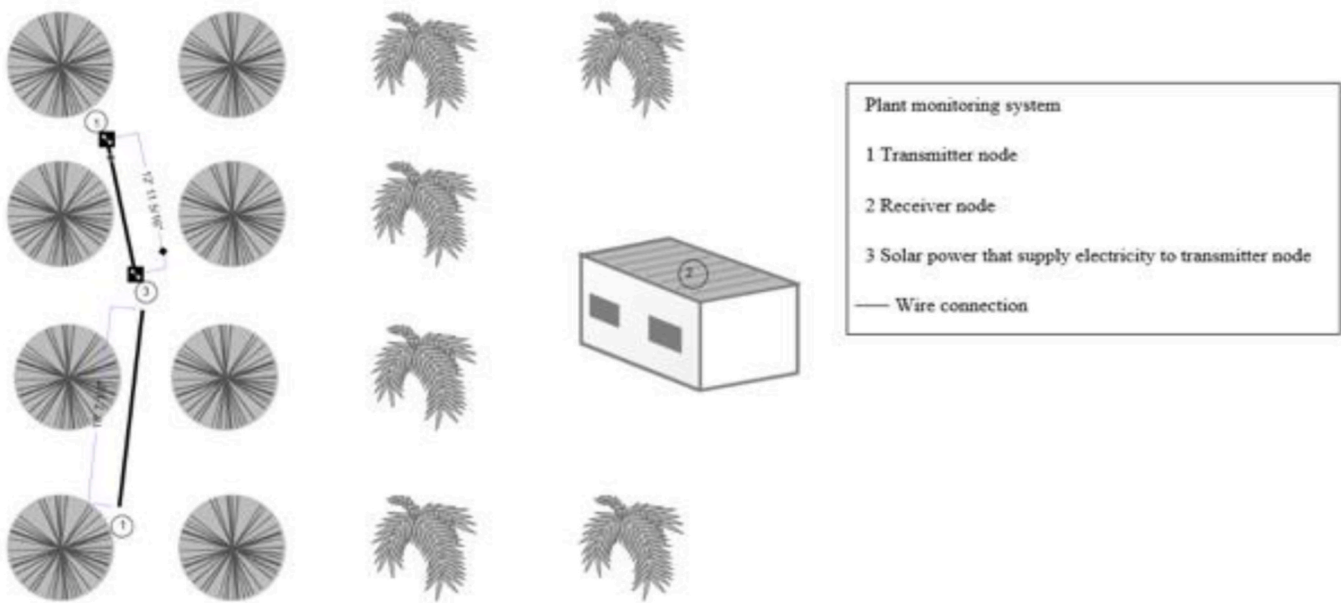


Fig. 4. Layout design for placement of the system in a fruit farm.

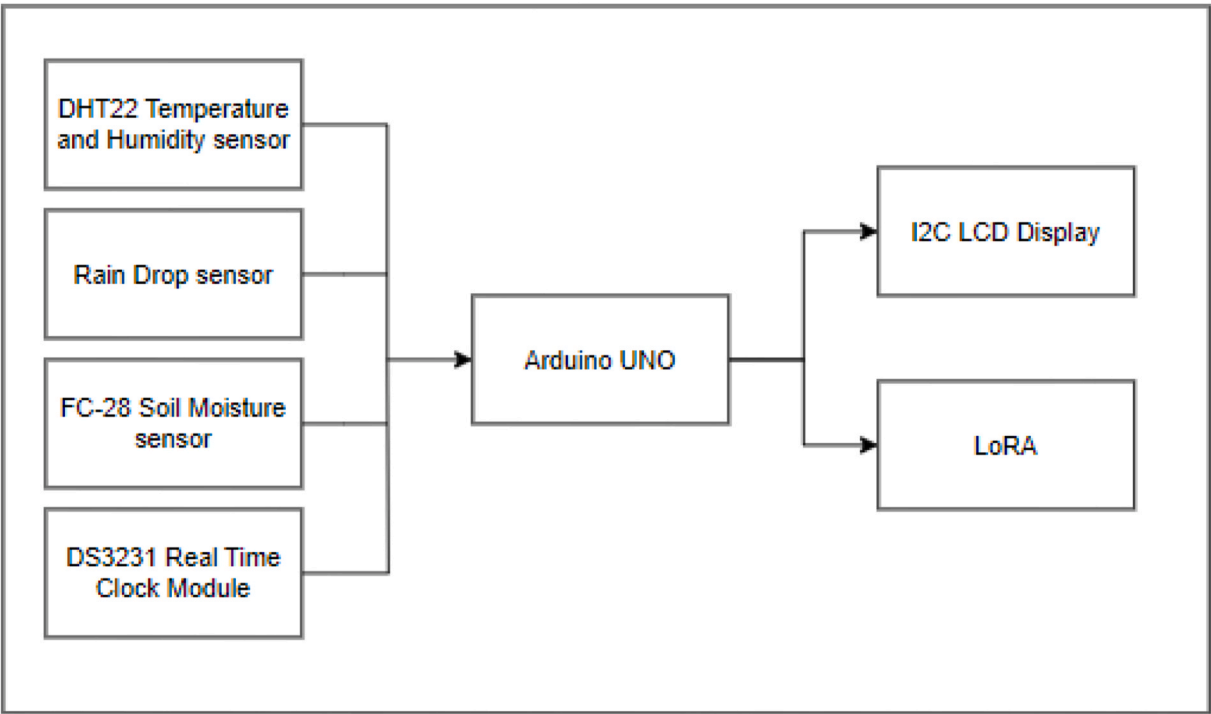


Fig. 5. Transmitter node sensing system block diagram.

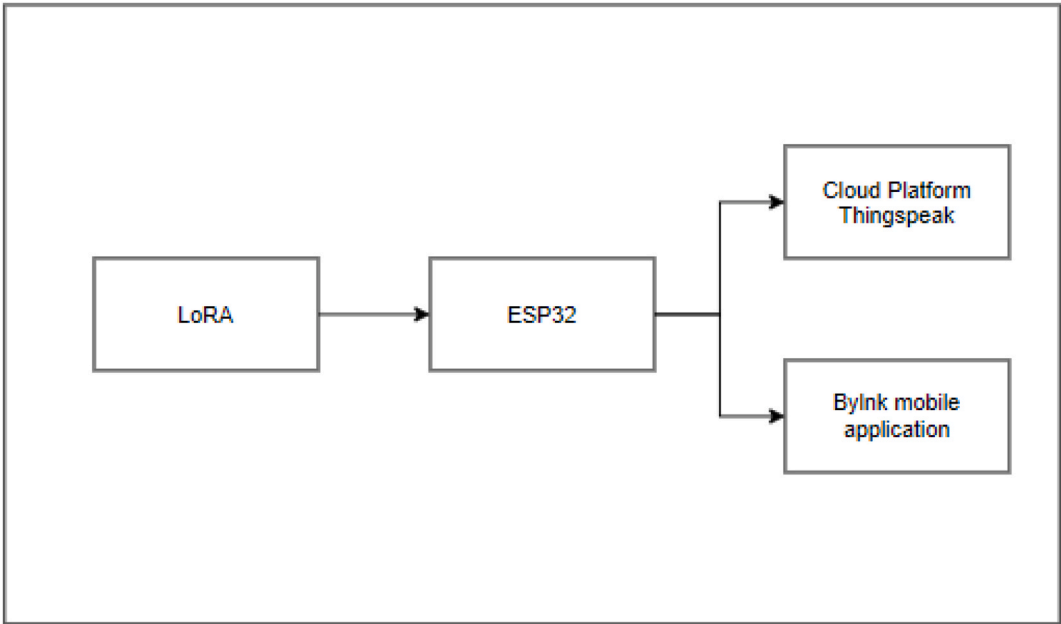


Fig. 6. Block diagram of receiver system.

transmission range of LoRa signals in an outdoor environment. A specific location with exact latitude and longitude coordinates was chosen for the receiver node (gateway module). The transmitter node was moved to several locations along the road, and the RSSI (Received Signal Strength Indicator) and SNR (Signal to Noise Ratio) readings were measured for each location, averaging three interval transmissions. Gradually increasing the distance, the test spanned a range from 0 to 923 m. These

results provide important information for enhancing LoRa-based sensor networks in smart farming applications, enabling better data gathering and monitoring in agricultural settings.

Conversely, at 0 m, the highest average RSSI of −59 dB and average SNR of 11.58 dB were recorded, indicating a strong signal with good quality. As the distance increased, both average RSSI and SNR values decreased, indicating signal degradation. Notably, the SNR values

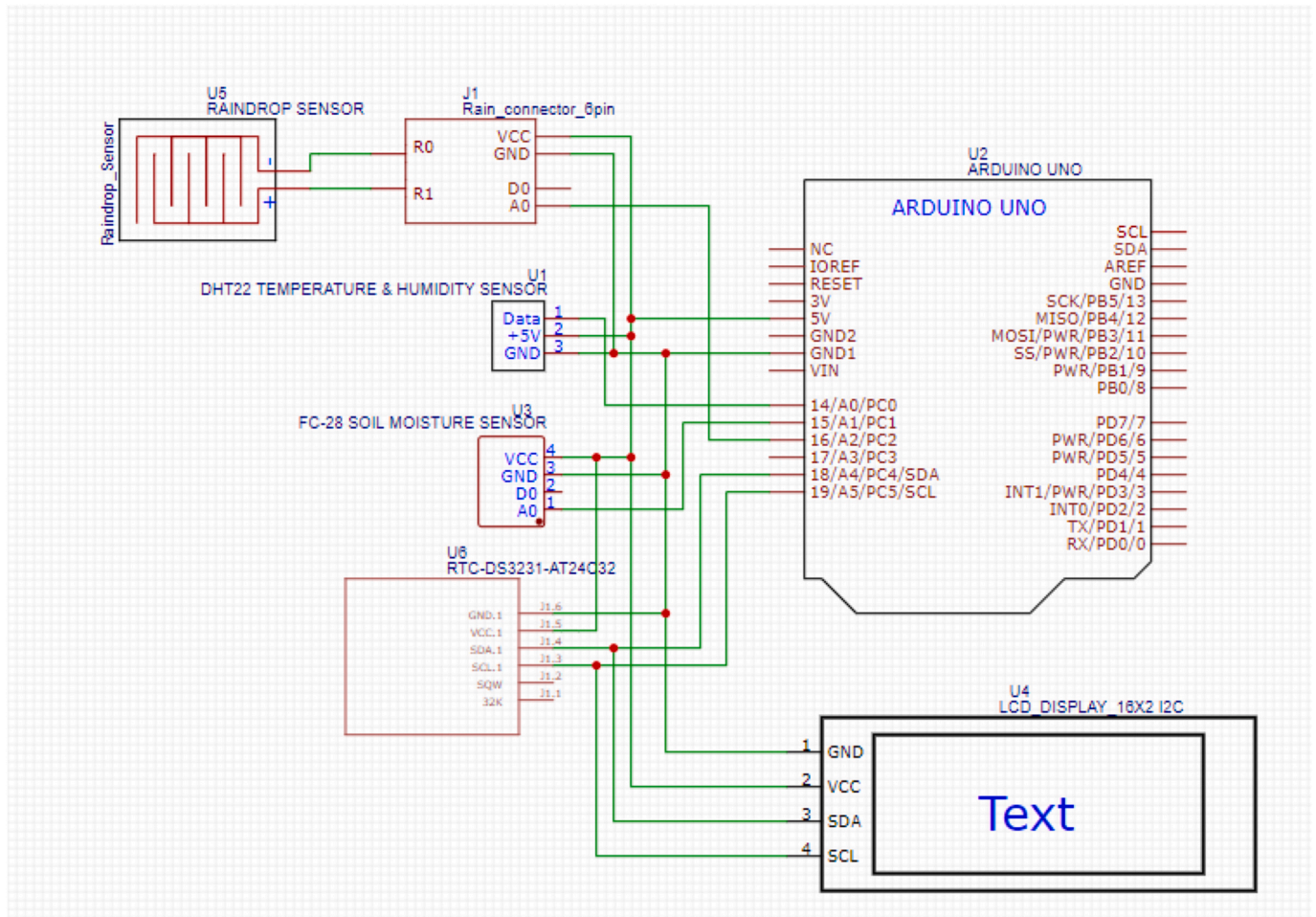


Fig. 7. Circuit connection for the transmitter node.

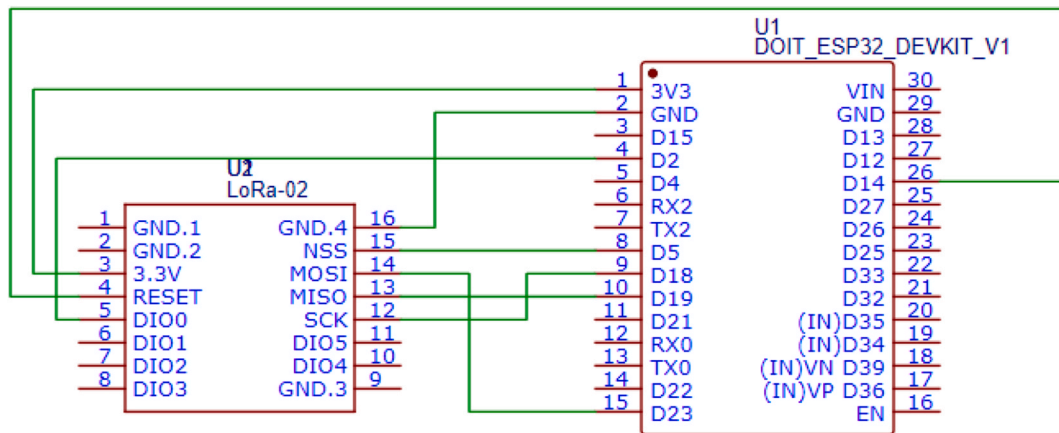


Fig. 8. Circuit connection for the receiver node.

indicated communication below the noise floor when the distance exceeded 80 m, suggesting dominant noise interference and compromised communication quality. For example, the average RSSI decreased to -86 dB ($p < 0.01$) at 20 m, which is a significant reduction from 0 m, yet the average SNR remained at 11.58 dB. When compared to longer distances, there was a statistically significant drop in both RSSI

(-113.67 dB) and SNR (1.83 dB) as the distance increased to 80 m ($p < 0.01$). SNR values below the noise floor were observed beyond 80 m, indicating a communication quality compromise caused by dominating noise interference. The lack of measurements above 800 m indicated a possible total loss of signal or signal-to-noise ratio, highlighting the LoRa communication system's limits at long ranges, a finding consistent with

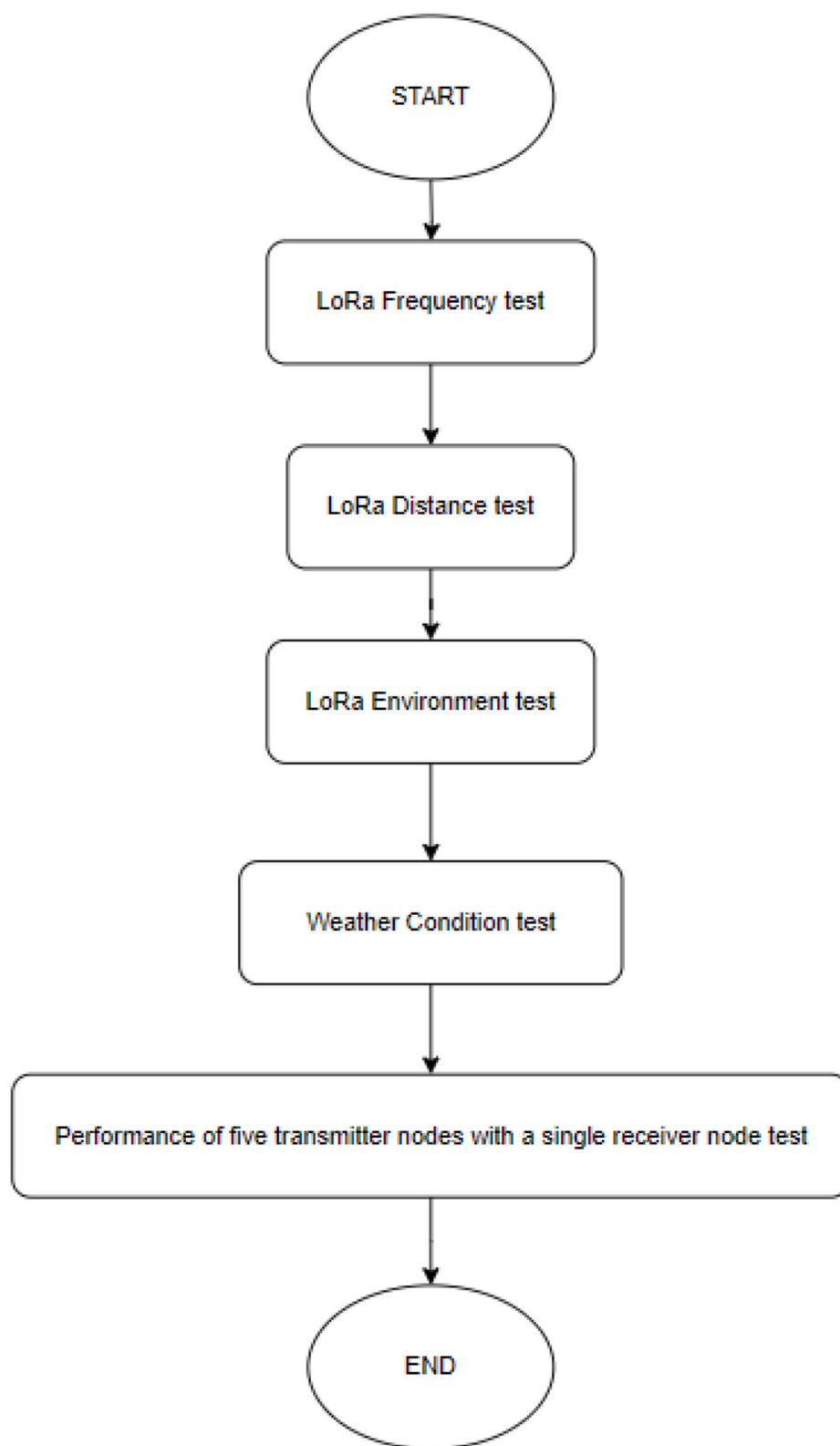


Fig. 9. Experimental flow chart overview.

Table 4
Experimental LoRa parameter setup.

Parameter	Transmit Power (dBm)	Value
Spreading Factor		12
Bandwidth (kHz)		250
Code Rate		4/5
Antenna Gain (dBi)		3

the observations in the work of [20] (see Table 6).

4.3. LoRa environment tests

Based on the results shown in Table 6, the LoRa distance test offered insightful information about the communication capabilities. With an average RSSI of -118.67 dB and an average SNR of -21.17 dB, the longest distance at which communication is reliably maintained

between the transmitter node and receiver node was found to be 700 m. However, after this distance, at 800, 900, and 923 m, no signal data was received, indicating that a solid link could not be made. Due to weak signal strength and low SNR, which hinder effective data reception, this is the case. For the LoRa environment test, four different environments were selected: a single-storey housing area, double-storey housing area, a high-rise building area, and a large farm area. The purpose of this test was to evaluate the radio frequency communication path when obstructed by various mediums, causing a loss of direct signal reception. Common obstacles that could fully or partially block signals included tall buildings, houses, trees, and areas with high voltage congestion. The test was conducted on a farm (Adap 4orty Farm) in Kidurong, Bintulu. In the residential areas, terrace houses were used as obstacles for the single-storey housing area and double-storey housing area tests, while a tall building complex was used for the high-rise building area test. In the farm area, a field of tall fruit trees was chosen as the obstruction.



Fig. 10. Blynk application monitoring display.

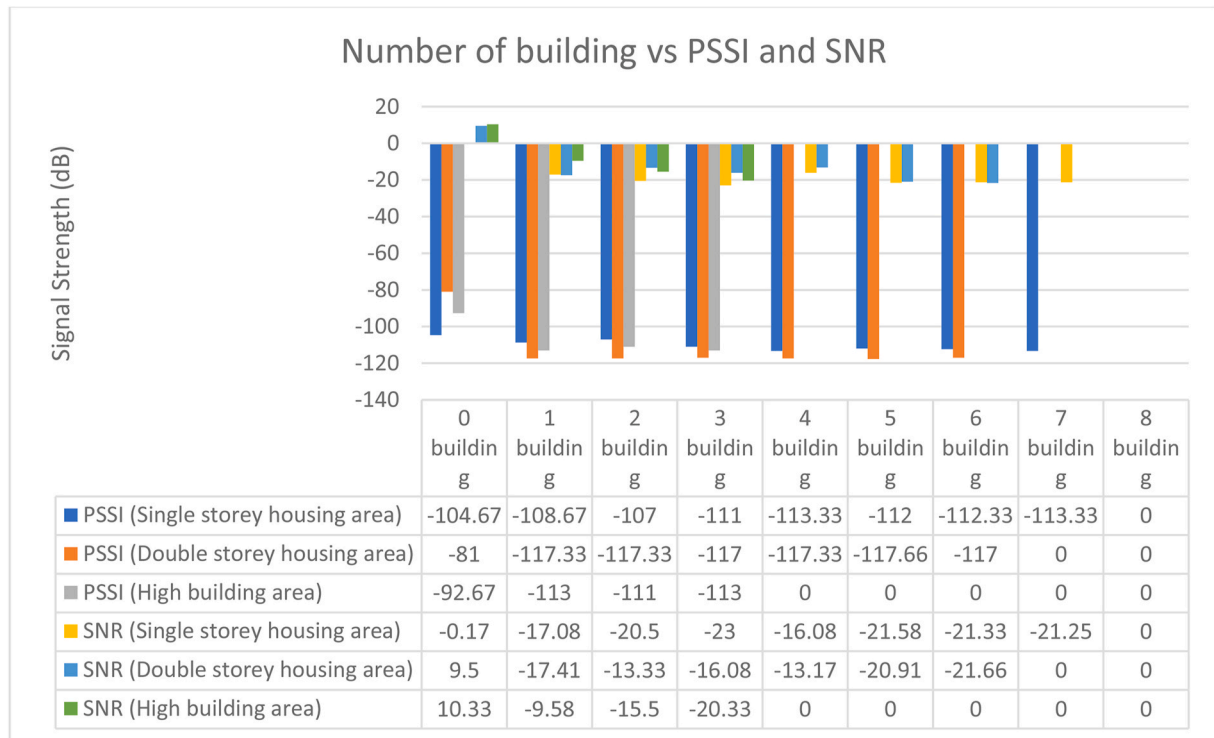
Table 5
RSSI and SNR Reading for LoRa Frequency test.

Frequency (Hz)	Distance (m)	PSSI (dBm)				Average PSSI (dBm)	SNR (dBm)			Average SNR (dB)
433	50	−95	−93	−89	−92		0.5	−2	8	6.50
	100	−111	−112	−112	−112		−6	−8	−7.5	−7.17
866	50	−121	−121	−121	−121		−23	−23	−18.75	−21.58
	100	−122	−122	−122	−122		−23	−23	−22	−22.67
915	50	−121	−121	−120	−121		−22.75	−18.25	−24.5	−21.58
	100	−122	−121	−122	−122		−23	−25	−24	−24

Table 6

PSSI and SNR Reading for LoRa distance test.

Distance(m)	RSSI (dBm)			Average RSSI (dBm)	SNR (dB)			Average SNR (dB)
0	-57	-60	-60	-59	13.25	10.25	11.25	11.58
20	-84	-87	-87	-86	10.75	12.75	12	11.17
40	-96	-95	-96	-95.67	9.75	8.5	9.75	9.34
60	-106	-107	-107	-106.67	6.25	7	6.75	6.68
80	-112	-115	-114	-113.67	3	0.5	2	1.83
100	-108	-107	-107	-107.33	6.25	6.25	6	6.17
200	-111	-111	-110	-110.67	4.25	3.5	3.25	3.67
300	-115	-115	-116	-115.33	-8	-7.25	-8.5	-7.92
400	-116	-115	-116	-115.67	-12	-13	-14.5	-13.17
500	-115	-117	-118	-116.67	-20.25	-17.75	-19.75	-19.25
600	-118	-118	-118	-118	-22.25	-20	-20	-20.75
700	-118	-119	-119	-118.67	-20.5	-22	-21	-21.17
800	nan	nan	nan	nan	nan	nan	nan	nan
900	nan	nan	nan	nan	nan	nan	nan	nan
923	nan	nan	nan	nan	nan	nan	nan	nan

**Fig. 11.** The graph of number of buildings vs. PSSI and SNR at single-storey housing areas, double-storey housing areas, and high-rise building areas.

According to Fig. 11, the single-storey housing area demonstrated the strongest LoRa signal strength compared to the double-storey housing area and high-rise building area. In the single-storey housing area, the average RSSI values ranged from -104.67 dB m to -113.33 dB m, exhibiting significant differences when compared to the double-storey housing area ($p \leq 0.01$) and the high-rise building area ($p \leq 0.05$). Similarly, the average SNR values ranged from -0.17 dB m to -21.25 dB m, showed statistical significance compared to both the double-storey housing area ($p \leq 0.01$) and the high-rise building area (p

≤ 0.01). The LoRa signal successfully penetrated a total of 7 block buildings in the single-storey housing area, 6 block buildings in the double-storey housing area, and 3 buildings in the high-rise building area. However, in this LoRa environment test involving 8 building cases, communication using the LoRa module failed due to signal distortion and attenuation, leading to blocked transmission. These findings highlight the negative impact of increased obstacles on LoRa coverage. Conversely, environments with fewer and smaller obstacles contribute to stronger wireless LoRa communication signals [21].

Table 7

RSSI and SNR result for farm area test.

Distance (m)	RSSI (dBm)			Average RSSI (dBm)	SNR (dB)			Average SNR (dB)
100	-106	-107	-107	-106.67	-15.25	-16	-17.25	-16.17
200	-107	-106	-107	-106.67	-15.75	-20	-20	-18.42
300	-106	-111	-112	-109.67	-21.75	-15.75	-19.5	-19
400	-114	-113	-114	-113.67	-19.50	-20	-19	-19.5
500	nan	nan	nan	nan	nan	nan	nan	nan
600	nan	nan	nan	nan	nan	nan	nan	nan

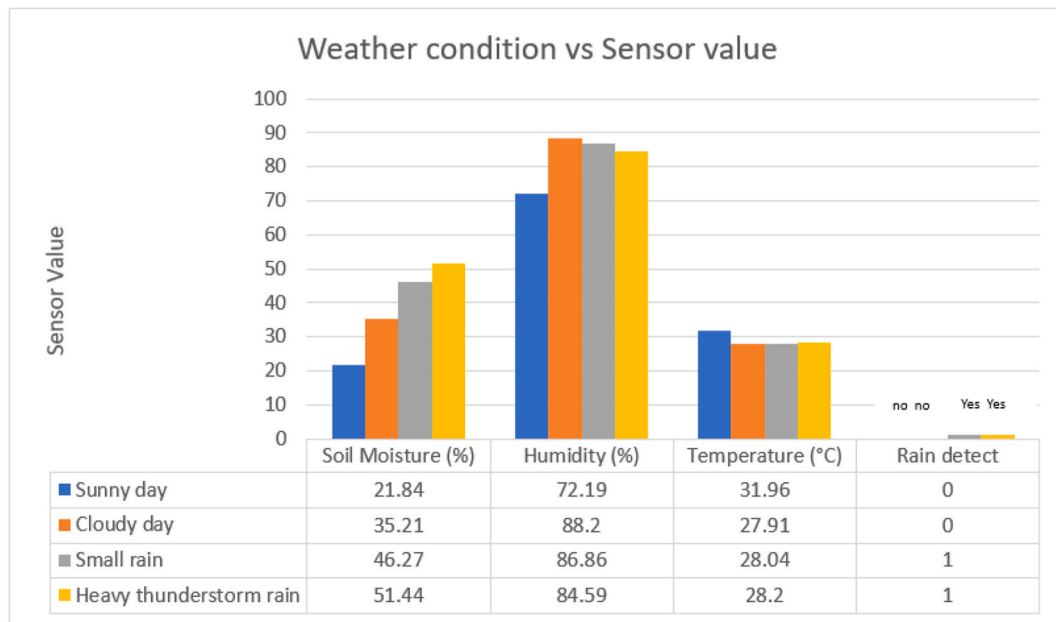


Fig. 12. The graph displays the average data for environmental temperature, humidity, soil moisture, and rain detection under various weather situations.

In Table 7, the farm area test conducted for LoRa communication in agricultural settings revealed valuable insights. The results indicated that LoRa signals achieved successful reception at 400 m, but beyond that range, the signal became too weak for reliable communication. The findings highlight the need to consider range limitations when deploying LoRa-based IoT solutions in large-scale farm areas. Specifically, the average RSSI values ranged from -106 dB m to -114 dB m within the effective range, showcasing a statistically significant difference in signal strength at different distances ($p \leq 0.05$). Beyond the 400-m mark, the decreasing trend in RSSI values underscores the diminishing signal strength. Despite the observed limitations, the results also underscore the potential benefits of efficient monitoring and automation within the achievable range, where LoRa signals maintain reliability and robust communication [22].

4.4. Weather condition tests

The weather test was conducted at a deserted area in Kidurong, Bintulu, Sarawak. The transmitter node and receiver node were positioned at predetermined locations. The nodes were spaced 50 m apart permanently. Data on soil moisture, humidity, ambient temperature, and rain detection were collected during the experiment under four different weather scenarios: sunny day, cloudy day, light rain, and heavy thunderstorm. To aid in the creation of integrated IoT solutions for smart farms, these tests aimed to offer insights into the effects of various weather conditions on the measured data.

The effect of various meteorological circumstances on soil moisture level, humidity level, environmental temperature, and rain detection is

shown in Fig. 12. The highest soil moisture level is produced by heavy thunderstorm rain, which is 51.44%. This indicates that the soil absorbs a significant amount of water during intense rain events. The lowest level is produced by sunny days, which is 21.84% ($p \leq 0.01$), indicating that evaporation and limited rainfall contribute to drier soil conditions. The lowest level of humidity, 72.19%, is seen on sunny days indicating drier air. The maximum level, 86.86% ($p \leq 0.05$), is seen during small rain. This suggests that the combination of moisture from rain and ambient atmospheric conditions increases humidity levels. On sunny

Table 9

Results of the packet transmission for five LoRa transmitter nodes in 1 h.

Round	Distance between receiver node and transmitter node				
	Five Transmitter Nodes with a Single Receiver Node				
	Node 1	Node 2	Node 3	Node 4	Node 5
	100 m	50 m	50 m	25 m	25 m
1	Received	Received	Received	Received	Received
2	Received	Received	Received	Received	Received
3	Missed	Missed	Received	Received	Received
4	Received	Received	Received	Received	Received
5	Received	Received	Missed	Received	Received
6	Received	Received	Received	Received	Received
7	Missed	Received	Received	Received	Received
8	Missed	Received	Received	Received	Received
9	Received	Received	Received	Received	Received
10	Received	Missed	Received	Received	Received
11	Received	Received	Received	Received	Received
12	Received	Received	Received	Received	Received

Table 8

Rounds of data transmission for five LoRa transmitter nodes within 1 hour.

Node	Round											
	Time in minutes when data sent by transmitter node 1 to 5											
	1	2	3	4	5	6	7	8	9	10	11	12
1	00	05	10	15	20	25	30	35	40	45	50	55
2	01	06	11	16	21	26	31	36	41	46	51	56
3	02	07	12	17	22	27	32	37	42	47	52	57
4	03	08	13	18	23	28	32	38	43	48	53	58
5	04	09	14	19	24	29	34	39	44	49	54	59

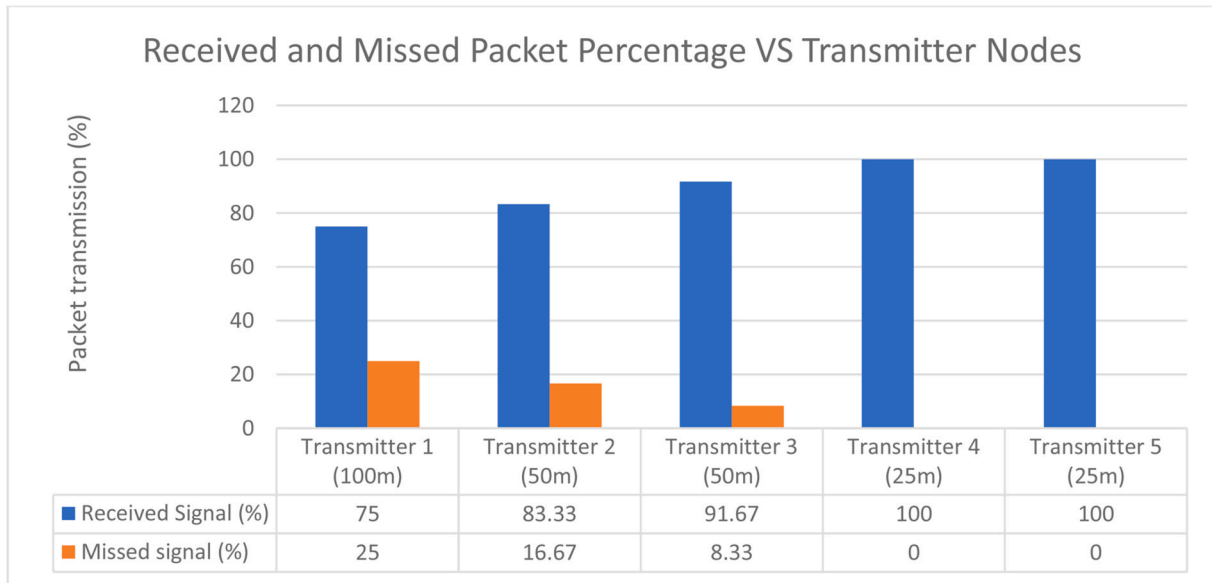


Fig. 13. The graph illustrates the received and missed packet percentages for five transmitter nodes signal transmission in 1 h.

days, the temperature dips to 27.91 °C due to reduced solar radiation, while on sunny days it reaches a maximum of 31.96 °C ($p \leq 0.01$). The direct exposure to sunlight and absence of cloud cover contribute to higher temperatures during the daytime. While no rain is detected on cloudy and sunny days, rain is sensed during heavy thunderstorm rain and small rain. This information is valuable for monitoring precipitation patterns and understanding the impact of rainfall on crop water requirements. By accurately detecting rainfall, farmers can make informed decisions regarding irrigation schedules and water conservation practises. The development of integrated IoT solutions for smart farms is aided by these findings, which offer insights into the meteorological conditions encountered in Malaysia's tropical environment and their effects on the measured metrics.

4.5. Performance of five transmitter nodes with a single receiver node

Five transmitter nodes coupled to one receiver device were tested over the course of an hour in this study to determine how well they performed. Five of the transmitter nodes had Arduino microcontrollers, and the other three used ESP32 microcontrollers, all of which were set up with the same sensor specifications. The receiver's performance was evaluated with a serial monitor and Blynk applications. To evaluate sensor capability and connectivity, the five transmitter nodes printed sensing data using the serial monitor and LCD Display. However, an issue arose where the receiver occasionally received random data when all the transmitter nodes transmitted simultaneously. A real-time clock module was installed in each transmitter node to solve this problem by synchronising the timing of data transmission. As shown in Table 8, this test evaluation consisted of 12 rounds of 5 min each, during which five transmitter nodes sequentially transmitted data for 1 min to a single receiver node. Table 9 displays the outcomes of the packet transmissions.

To have a better understanding, the accumulated data was transferred into a percentage value using Eq. 1 and converted into a graphical chart for better visualisation and data interpretation, as shown in Fig. 13 below [23].

$$\text{Packet Loss(\%)} = \frac{\text{Total Number of Packet Loss}}{\text{Total Number of Packet}} \times 100\% \quad (1)$$

The analysis of the graph presented in Fig. 13 provides valuable insights into the performance of the five transmitter nodes over a 1-h signal transmission period. Among the transmitter nodes, Transmitter node 1 displayed the highest percentage of missed signals, reaching 25%, suggesting potential issues with signal interference or weak signal strength. This can be attributed to Transmitter node 1 being located at the longest distance from the receiver, spanning 100 m. Statistical analysis confirmed the significance of the observed differences, emphasizing the impact of distance on signal reliability ($p \leq 0.01$). Transmitter node 2 followed with a missed signal rate of 16.67%. Meanwhile, Transmitter node 3 exhibited a moderate missed signal rate of 8.33%, indicating relatively stable transmission. On the other hand, Transmitter nodes 4 and 5 performed exceptionally well, with no missed signals owing to their proximity to the receiver node, situated at 25 m. Statistical tests supported the significance of the superior performance of Transmitter nodes 4 and 5 compared to others ($p \leq 0.01$).

These findings underscore the critical importance of signal reliability in LoRa-based, IoT-enabled wireless sensor networks for smart agriculture. Addressing signal quality issues, especially for transmitter nodes located at longer distances, through optimisation and signal amplification can significantly enhance data transmission performance, facilitating more informed decision-making for farmers and supporting sustainable agricultural practises.

4.6. Discussion of limitations

4.6.1. Hardware limitations

The implementation of IoT-enabled smart farm systems encounters several hardware limitations. Availability and compatibility issues with hardware components, such as sensors and microcontrollers, may restrict options and functionality. Inadequate processing power and memory capacity in selected devices can lead to delays and reduced system responsiveness, impacting real-time data processing. Power consumption considerations, especially in remote environments, necessitate energy-efficient components and power-saving strategies. Connectivity challenges arise due to the vast and varied landscapes of smart farms, affecting data transmission and remote-control capabilities. Durability and ruggedness are crucial in agricultural settings, where IoT devices are exposed to harsh conditions. Building a large-

scale LoRaWAN network for agriculture requires powerful gateways and strategic placement to ensure continuous and reliable connectivity, but the associated costs and logistics pose challenges. Overcoming these hardware limitations involves careful planning, selecting robust components, and implementing energy-efficient and resilient network structures to deliver valuable insights for optimized farm management.

4.6.2. Software limitations

The software limitations in the implementation of the smart farm system primarily revolve around the constraints of the Bylnk application and its integration with the Arduino IDE. While Bylnk offers various features, its limitations in customization and extensibility for basic users may impact the adaptability to smart farm requirements. The interface and widgets, tailored to general IoT applications, may not fully align with the specific needs of smart farms, affecting user experience and customization options. Integration with the Arduino IDE poses challenges in advanced debugging, code optimisation, and scalability, particularly for large-scale deployments. Overcoming these limitations involves exploring alternative software platforms, implementing modular design, and adhering to software engineering best practices to ensure optimal utilization of Bylnk and Arduino IDE while meeting agricultural domain requirements.

5. Conclusions

The “Optimising Performances of LoRa based IoT Enabled Wireless Sensor Network for Smart Agriculture” project successfully demonstrated the feasibility and potential of integrating IoT technologies in agricultural settings. The adoption of LoRa technology in smart agriculture brings significant advancements to traditional data collection and analysis methods within the agro-environmental system [24]. By implementing a robust system that combines sensors, microcontrollers, and mobile applications, the project enabled real-time monitoring and management of critical environmental parameters for optimising crop growth. Through extensive testing and evaluation, valuable insights were gained into the system’s performance and functionality, identifying areas for improvement in both hardware and software components. While the project achieved its objectives, there is room for further development, such as enhancing scalability, improving the user interface, and integrating additional functionalities. The project contributes to the ongoing effort of creating smarter and more interconnected farms, with the potential to enhance efficiency, productivity, and sustainability in farming practises. The key findings from experimental results, such as the optimal LoRa frequency and maximum transmission distance, further inform the implementation of IoT technologies in agriculture. Leveraging IoT systems in smart farms can empower farmers with real-time data for data-driven decision-making, leading to improved resource management, sustainability, and crop production. The scalability and adaptability of the developed system open avenues for future innovation, including advanced analytics and machine learning algorithms. This transformative approach aims to evolve traditional farming practices into technologically advanced, efficient, and sustainable systems, with broader implications and applications across diverse agricultural landscapes.

CRedit authorship contribution statement

Yik-Tian Ting: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Kah-Yoong Chan:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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